Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Vibrational spectroscopy(harmonic oscillator)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Vibrational spectroscopy(harmonic oscillator)

1 Vibrational spectroscopy(harmonic oscillator)

Study focus: Vibrational spectroscopy(harmonic oscillator). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000066 · EV0000067

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module one on molecular spectroscopy.

Key details

- And in today's class we are going to study about vibrational spectroscopy.
- We will tie try to derive an expression for the vibrational energy levels that is for a harmonic oscillator and also we will arrive at the vibrational spectrum for a harmonic oscillator.
- Like I have explained this before when we were talking about the electromagnetic spectrum.
- It is vibrating.
- And then compresses stretches and compresses that would result in the a vibrational change in the molecule.
- Now when there is a vibration in the molecule, then the dipole moment is also changing because we know HCl is a polar molecule you have a slight positive charge and a slight negative charge.
- The direction remains same but same but the magnitude is changing.
- Field is also changing.
- The electric field now is there is a frequency with which it is vibrating, and therefore the electric field will also fluctuate with a particular frequency.
- We know that electromagnetic radiation it has an electric field as well as a magnetic field and based on the energy of the radiation, the electric component of the electromagnetic radiation is also going to be fluctuating.
- One vibrational level to the next vibration level.
- Rotational spectroscopy we know that we use the microwave energy.
- But here we are going to use the IR radiation.

- That means that the vibrational energy levels, the energy gap of these vibrational energy levels matches with the energy gap of the IR radiation.
- Because only when these frequencies there is going to be a resonance phenomena and the molecule goes from the lower vibrational level to the next vibrational level.
- So what we see as a result of this transition, because there is a transition which is happening from the of the molecule from lower vibration level to the higher vibration levels, we will see that a particular energy is being absorbed from that whole IR region beam.
- So therefore that would if we can record.
- Is that energy that is getting absorbed?
- Then we can get what we call as the vibrational spectrum.
- And once we have the vibrational spectrum with us, we will be able to understand about the nature of the molecule and even.
- Then we can identify what is the molecule.
- Therefore, this vibrational spectroscopy is like a fingerprint for the molecule.
- These are examples.
- Why these examples?

Example or application from the lecture

- Suppose we have an H Cl molecule and it is going to undergo.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 05:00–09:59

They there when the vibration takes place there has to be a change in the dipole moment.

Key details

- So therefore these molecules we know are polar in nature and therefore we will find that they will vibr when they vibrate there is going to be a change in the dipole moment.
- But when we.
- Come to homonuclear diatomic molecules like H2 and Cl2, we observe that these will not be active in the IR region.
- Because they do not possess a dipole moment.
- They are found to be I inactive.
- Can we say that if a molecule possesses permanent dichole moment, then it will be active in the IR region, just like we said about rotational spectroscopy?
- Here there is a small difference that CO2 molecule, we know that.
- It is microwave inactive because it does not possess a permanent dipole moment.
- But here in this case, in the IR region, we find that CO2 molecule is actually IR active.
- The reason is.
- Here, the criteria for observing a transition between the vibration levels is that.
- When the molecule is vibrating, there should be a change in the dipole moment, and that happens in the case of the CO2 molecule.

- Though it doesn't possess a permanent dipole moment, we know that the CO2 molecule, each of these bonds possess a dipole moment which actually cancel against each other, but when it vibrates we will find that in certain modes of vibration the CO2 molecule we can actually see a change in the dipole moment.
- But if the molecule is undergoing asymmetrical stretching, that is you can see one bond is becoming small, one bond is stretching, and then it goes in the reverse direction.
- If this happens, then we will find that there is definitely a change in the type moment because these two moments are not getting cancelled.
- So therefore we find that in this mode it becomes IR active.
- Because there is a change in the diple moment.
- And then if it is bending, say in the plane it is bending like this, again we see there is a change in the diple moment, these movements are not getting cancelled now, and therefore in the bending mode also we find that it is I active.
- And we will arrive at the energy expressions and see.
- why is the molecule able to show the vibrational spectrum.
- And again as we did in rotational spectroscopy we need a model so that we can apply that model and get to the energy expression.
- We are taking a heteronuclear diatomic molecule, so one mass will be larger than the other.
- And now we have a spring which is attaching the two atoms or holding the two atoms.
- So therefore when the molecule stretches the spring is getting stretched, when it is compressing the spring is getting compressed, and in the equilibrium position it remains like this.

Example or application from the lecture

- in the case of symmetric stretch, now suppose the CO2 is stretching symmetrically about the center then we about its equilibrium position we will find that there is no change in the dipole moment at all.

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Core explanation · 10:00–14:59

And since we it is a simple harmonic motion that we observe and it's a simple harmonic oscillator we can say that the more this vibration is going to obey hook's law.

Key details

- These molecules are never at rest.
- These bonds are always vibrating in different ways.
- Therefore, if we can get the IR spectrum, then we will get a fingerprint of the molecules.
- And if that will be very unique to the molecule.
- It obeys Hook's law, simple harmonic oscillator, f is equal to minus of kx.
- Any displacement is now related to the force by minus of kx.
- R minus r equilibrium will be the displacement.
- This means that K would basically be F.

- By X, ignoring the negative sign because it's just because it's a restoring force.
- So if the spring constant or the force constant is very high, that means even for a small distance.
- We need to apply a large amount of force.
- And looking at this expression we can understand that the energy curve is going to be parabolic.
- what is the reason why we get energy profile like this?
- As a rule, we understand that the heavier atoms we are going to fix at a particular position and it is the lighter atom which is going to be moving when the molecule stretches.
- That means we are trying to expand it.
- H now we are stretching.
- Earlier this was the bond length, you are providing more energy, you are stretching it more, so the bond length increased and it will reach this position, and when you compress it, it comes back to this position again, both are of higher energy.
- Every one has a.

Example or application from the lecture

- So let us say that the Cl is fixed because being the heavier and since it is at equilibrium so at this is the equilibrium distance or the internuclear distance at Cl that is in the equilibrium position.

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Core explanation · 15:00–19:59

M1 plus M2.

Key details

- So if we want to express in terms of wave number because we are interested in spectroscopic units, so nu bar oscillation would be 1 by 2 pi.
- C root of K by mu.
- 1 by 2 pi C root of K by mu centimeter inverse.
- Mu is a known quantity for a particular molecule.
- K, what is K?
- It is known as a force constant, but in the case of a molecule, what would be K?
- K would represent the bond strength.
- Then somehow, if I can find out what is the new bar oscillation for this harmonic oscillator, I will be able to get the value of K using this expression.
- Now what we do just like we did in the case of rotational spectroscopy we need to solve the Schrödinger equation because so far we are talking of classical mechanics but now we know we are talking of a molecule we cannot apply classical mechanic laws mechanics laws therefore we need to you go give it a quantum treatment.
- That is in Joule.

- We got a term V which is known as which is the vibrational quantum number and can take up values zero upwards.
- That is V is equal to 0, 1, 2, 3, and so on so forth.
- That means what this energy now expression because of the V has become quantized.
- But here now the energies are have become discrete or they have become quantized.
- And when we do it in wave numbers we know that it's represented by this small e and this is equal to v plus half nu bar centimeter inverse.
- So here it was V plus half h nu but because we are converting it dividing by H C, so this becomes nu bar centimeter.
- inverse so this is an important expression and this is going to give us the various energy levels when we start substituting v is equal to 0123.
- So we got that expression when we substitute V is equal to 0 because it was V plus half nu bar.
- If we substitute V is equal to 1, then we get 3 by 2 mu bar centimeter inverse.
- If you substitute 01 to V, I can get the vibrational levels in the molecule.
- What it looks like.
- The energy levels like this.
- In the classical mechanics, all these energies are allowed, but now we see that the energy of the quotient.
- What is the energy?
- Hal new bar.
- So that means it is not here, but it has a value that is half new bar, and therefore this energy is known as zero point energy.
- What does it mean that.
- The lowest energy level the lowest energy level is not at an ease equal to zero.
- We are not getting ease equal to zero, rather we are getting a value for this and that is half new bar.
- This is referred to as zero point energy.
- That is three by one.

Example or application from the lecture

- Let us see where these energy levels are appearing.
- Basically means that a molecule is never addressed, but it is always vibrating with energy which is known as zero energy and given by now let us start two B is equal to one.

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Core explanation · 20:00–24:59

two energy scale so let us go to the next one and that will be three by two new bar so now we have the next energy level now when we so this is zero this is one now let us say we have two here again this is five by two new bar and we have a four year save that will be three.

Key details

- And then it would be seven by two nubar.
- We get half nubar, three by two nubar, five by two nubar, and seven by two nubar.
- thing is we are interested in the energy gap because that will actually tell us the absorption that is going to take place.
- Between zero and one.
- What is the difference in energy?
- That would be B is equal to one.
- Minus V is equal to zero.
- V is equal to one is substitute energy expression.
- And then V is equal to zero is substrate the energy expression and V get the energy gathered.
- And that would be three by two minus half.
- new bar and that is equal to new bar.
- So therefore between these two energy levels 0 to 1, the energy gap is new bar.
- That is one to two.
- Again you are five by two minus three by two are so and again when you go from two to three again the energy differences so what do we understand here that energy levels.
- The quantity of energy levels difference is equal to nu bar oscillation centimeter inverse.
- where the energy levels are.
- You also know what is the energy gap.
- is still not sufficient for me to arrive at the spectrum.
- What is important, I should know what are the selection rules.
- When it remains the same that it is same for all molecules.
- That when a molecule vibrates there should be a change in the diagonal moment.
- And what is the specific selection rule for the simple harmonic oscillator diatomic molecule?
- It says that Dell B is equal to plus minus.
- What does it mean?
- That the only transition allowed will be zero to one or one to two or two to three.
- You cannot have zero to two or one to three and or any of them any other transition two to four, etc.
- Therefore we see that now zero to one where would the line appear.
- What do you mean by where would the line appear in the spectrum?
- That it is going to absorb that much energy equal to the energy gap that we have.
- From V is equal to one, V is equal to two, again the entity are the same.
- It is equal to number centimeter in now we can see from the spectrum here.
- from the rotation the vibrational levels you can see that the energy gap is constant.
- And this is.

Example or application from the lecture

- Let us go from 3 by 2 to 5 by 2.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 25:00–29:59

being restributed by L V is equal to plus minus one.

Key details

- Therefore when you see the absorption you can only go to the next constituent level.
- Therefore the spectrum is going to look like this.
- Just one line is observed in the spectrum and that will be at new bar centimeter in.
- The other terms are known to me.
- from the spectrum.
- And this is going to.
- Help us find out what is the bond strength of the molecules.
- Rotational spectroscopy saw that it gives us the bone length and vibrational spectroscopy is going to give us the bone strength.
- So in the next class we are talking to talk about how this harmonic oscillator model is not suitable for molecules, and we will take up [ASR low confidence; check audio].
- Vibrational spectroscopy by changing the model to an end-harmonic oscillator.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Vibrational spectroscopy(harmonic oscillator) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Vibrational spectroscopy(harmonic oscillator) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(harmonic oscillator) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/005_Vibrational_spectroscopy_harmonic_oscillator__bd1d3b496879b403.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00015	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00017	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

C Revision and retrieval practice